Week 4-2

Discrete Random Variable: Expected Value

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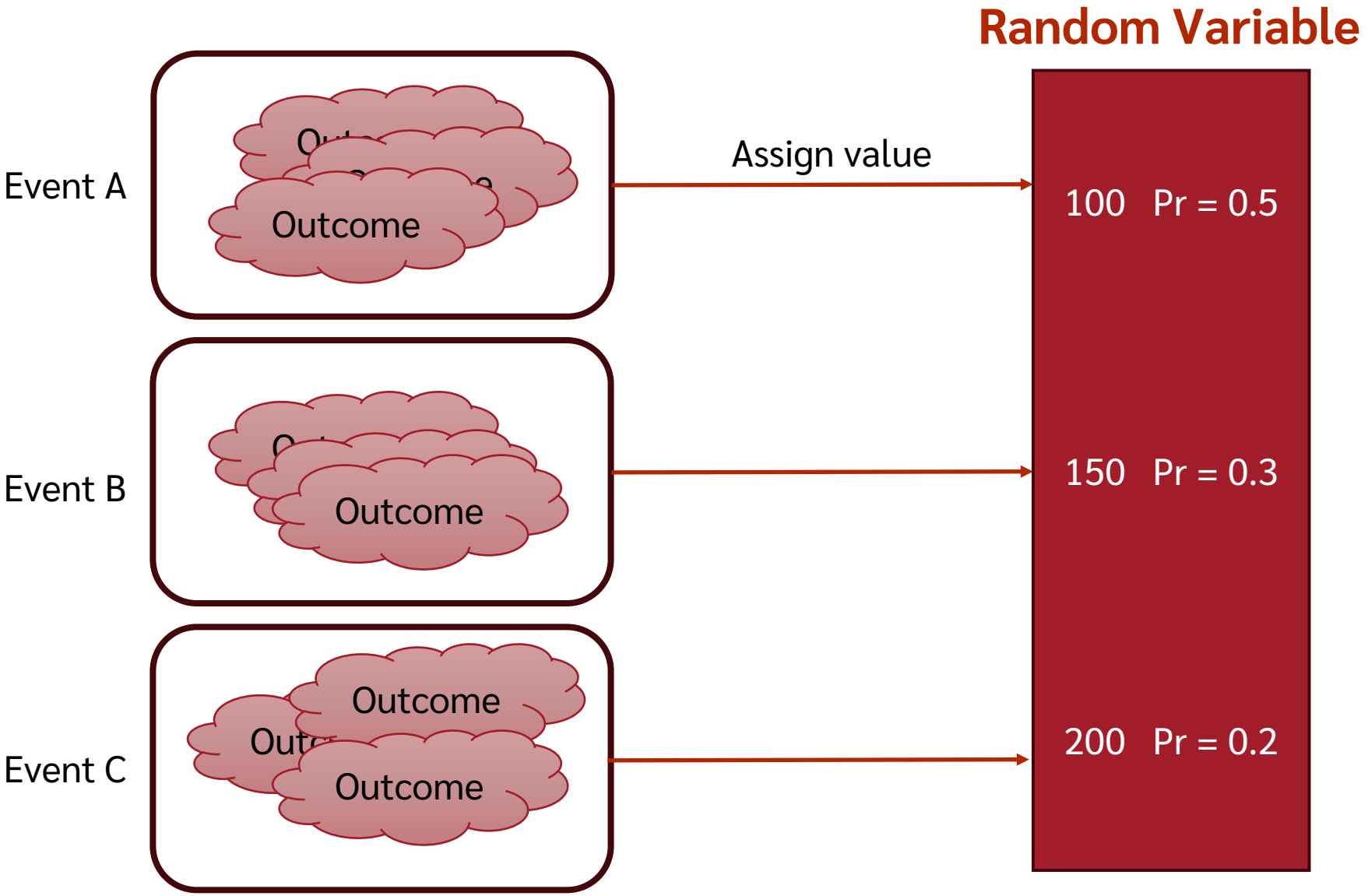
2110205 STATS COMP ENG

CHULA
COMPUTER
ΣNGINEERING

Recap

Discrete Random Variable



Value DEPEND on the outcome.

Arithmetic in Random Variable

Arithmetic

- **Addition** $\mathbf{Z = X + Y}$
- **Subtraction** $\mathbf{Z = X - Y}$
- Multiplication $Z = X \times Y$
- Square $Z = X^2$

Attendance Check

Supposed:

- $X_1 \sim Bernoulli(0.5)$
- $X_2 \sim Bernoulli(0.5)$
- $X_3 \sim Bernoulli(0.5)$

What would be the distribution of $X_1 + X_2 + X_3$?

Addition

Recall: Binomial is just multiple Bernoulli trials.

$$Bin(3, p) = Ber(p) + Ber(p) + Ber(p)$$

$Ber(p)$ is 0 or 1 for a single trial. Summation of them is the number of success for n trials.

Arbitrary addition

Example: Suppose X and Y has the following pmf.
Find pmf of $X + Y$.

X	1	2	3
p(a)	**1/6**	**2/6**	**3/6**

Y	1	2	3	4
p(a)	**1/10**	**2/10**	**3/10**	**4/10**

Step 1: Draw a table of the sample space.

Step 2: Sum up all possible combination.

Table of sample space

X \ Y	1 Pr=1/10	2 Pr=2/10	3 Pr=3/10	4 Pr=4/10
1 Pr=1/6	Z = 2 $Pr = \frac{1}{6} \times \frac{1}{10} = \frac{1}{60}$	Z = 3 $Pr = \frac{1}{6} \times \frac{2}{10} = \frac{2}{60}$	Z = 4 $Pr = \frac{1}{6} \times \frac{3}{10} = \frac{3}{60}$	Z = 5 $Pr = \frac{1}{6} \times \frac{4}{10} = \frac{4}{60}$
2 Pr=2/6	Z = 3 $Pr = \frac{2}{6} \times \frac{1}{10} = \frac{2}{60}$	Z = 4 $Pr = \frac{2}{6} \times \frac{2}{10} = \frac{4}{60}$	Z = 5 $Pr = \frac{2}{6} \times \frac{3}{10} = \frac{6}{60}$	Z = 6 $Pr = \frac{2}{6} \times \frac{4}{10} = \frac{8}{60}$
3 Pr=3/6	Z = 4 $Pr = \frac{3}{6} \times \frac{1}{10} = \frac{3}{60}$	Z = 5 $Pr = \frac{3}{6} \times \frac{2}{10} = \frac{6}{60}$	Z = 6 $Pr = \frac{3}{6} \times \frac{3}{10} = \frac{9}{60}$	Z = 7 $Pr = \frac{3}{6} \times \frac{4}{10} = \frac{12}{60}$

X+Y	2	3	4	5	6	7
p(a)	**1/60**	**4/60**	**10/60**	**16/10**	**17/60**	**12/60**

What about subtraction?

- **Subtraction?**
  - $X - Y$ is the same as $X + (-Y)$ so just flip the sign on Y.
- **Multiplication?**
  - Getting the possible value of each Z is easy, but calculating pmf is extremely hard and outside of the scope for now.

Expected Value

Expected Value

- The average value of the **Random Variable**.
- Denoted by $E(X)$ or μ.

Example 1

Example: Average of throwing a 6-sided dice 6,000 times?

Answer: Let do it.

```
[1] import random
```

```
[2] result = []
    for i in range(6000):
      result.append(random.randint(1,6))
    result[:10]
```

```
[2, 3, 2, 5, 1, 3, 1, 5, 4, 6]
```

```
[3] avg = sum(result) / len(result)
    avg
```

```
3.4905
```

Probably around 3.5

Example 1 (cont'd)

Or, we can try to calculate it.

Let X be the random variable for the face of the dice.

X	1	2	3	4	5	6
p(a)	**1/6**	**1/6**	**1/6**	**1/6**	**1/6**	**1/6**
Expected count	1000	1000	1000	1000	1000	1000

The average is $\frac{1\times1000+2\times1000+3\times1000+4\times1000+5\times1000+6\times1000}{6000} = 3.5$

Example 2

Example: We have a fair dice. You win $1000 if the dice is 1 or 2. You lose $300 if the dice is 3, 4, 5, 6. Do you play the game?

X	**$1000**		**-$300**			
Dice face	1	2	3	4	5	6
	1/6	1/6	1/6	1/6	1/6	1/6
p(a)	**2/6**		**4/6**			

Expected Return: $\$1000 \times \frac{2}{6} - \$300 \times \frac{4}{6} =$ **$133.333**

Back to Horse Racing

Who should we bet on?

Calculate the expected value:

$$100 \times 0.5 + 150 \times 0.3 + 200 \times .2 = 135$$

Since it is a positive, we would bet on..... ?

WRONG!

Who wins	Payout	Prob.
A	100	0.5
B	150	0.3
C	200	0.2

Back to Horse Racing

We don't even know how much to bet.

Assume it cost 50 to bet.

To bet on A: $(-50) + 100 \times 0.5 = 0$

To bet on B: $(-50) + 150 \times 0.3 = -5$

To bet on C: $(-50) + 200 \times 0.2 = -10$

ANSWER: We should not bet on anyone.

Who wins	Payout	Prob.
A	100	0.5
B	150	0.3
C	200	0.2

Quiz 1

1. What is the distribution of $Binomial(n, p) + Binomial(m, p)$? Why?

2. Calculate $E(X)$ when $X \sim Bernoulli(0.8)$.

10 MINUTES BREAK

The Definition

Definition: X is a discrete random variable with value $x_1, x_2, \ldots, x_n$ with probability $p(x_1), p(x_2), \ldots, p(x_n)$. The expected value $E(X)$ is:

$$E(X) = \sum_{j=1}^{n} p(x_j) x_j.$$

Example 1

Example: Let $X \sim Bernoulli(p)$. Find $E(X)$.

Answer:

$$E(X) = \sum_{j=1}^{n} p(x_j)x_j$$

$$= 0 \times (1-p) + 1 \times p$$
$$= p$$

Algebra with E(X)

Properties 1: $E(X+Y) = E(X) + E(Y)$ if X and Y is on the same sample space.

Properties 2: $E(aX+b) = aE(X) + b$ if a and b is constant.

Example 2

Example: Roll 2 dice. X is the sum of the value. Find $E(X)$.

Answer: 7.

X	2	3	4	5	6	7	8	9	10	11	12
p(a)	**1/36**	**2/36**	**3/36**	**4/36**	**5/36**	**6/36**	**5/36**	**4/36**	**3/36**	**2/36**	**1/36**

OR: Let X_1 be the value of the 1st dice, and X_2 be the 2nd dice.

We know $E(X_1) = E(X_2) = 3.5$, so $E(X) = E(X_1) + E(X_2) = 7$

Example 3

Example: Let $X \sim Binomial(n, p)$. Find $E(X)$.

Answer: Let $X_j \sim Bernoulli(p)$. We have previously seen that:

$$X = \sum_{j=1}^{n} X_j$$

So:

$$E(X) = \sum_{j=1}^{n} E(X_j) = \sum_{j=1}^{n} p = np.$$

Example 4

Example: Let X be a random variable with infinite value.

X	2	2^2	2^3	…	2^k
p(a)	$1/2$	$1/2^2$	$1/2^3$	…	$1/2^k$

What is $E(X)$?

Example 5

Example: What about Geometric Distribution?

X	0	1	2	…	k
p(a)	p	$(1-p)p$	$(1-p)^2p$	…	$(1-p)^kp$

Answer: $E(X) = \sum_{k=0}^{\infty} k(1-p)^k p = \frac{1-p}{p}$

How?

Example 5 (cont'd)

Sum of the geometric series: $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$

Take $\frac{d}{dx}$ to both side: $\sum_{k=0}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}$

Multiply by x: $\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2}$

Replace x by $1-p$ $\sum_{k=0}^{\infty} k(1-p)^k = \frac{1-p}{p^2}$

Multiply by p $\boxed{\sum_{k=0}^{\infty} k(1-p)^k p} = \frac{1-p}{p}$

What we wanted.

Example 6

Example: The probability that you pass this class is 0.9. On average, how many time you would fail before passing this course?

Answer: Let $X \sim Geometric(0.9)$.

Then, $E(X) = \frac{1-p}{p} = \frac{1-0.9}{0.9} = \boxed{0.111\ldots}$

Example 7

Example: A lootbox without guarantee has a drop chance of 0.05.

How many times you need to open the box?

Answer: Let $X \sim Geometric(0.05)$.

Then, $E(X) = \frac{1-p}{p} = \frac{1-0.05}{0.05} = \frac{0.95}{0.05} = \boxed{19}$

Change of Variable Formula

If X is a discrete random variable, and h is a function, then $\boldsymbol{h(X)}$ **is a new random variable.**

$$E\big(h(X)\big) = \sum_j h\big(x_j\big)p(x_j)\,.$$

WARNING: $E\big(h(X)\big) \neq h\big(E(X)\big)$!

Example 8

Example: Let X be a random variable for value of one dice roll. Let $Y = X^2$. Find $E(Y)$.

Answer:

X	1	2	3	4	5	6
Y	**1**	**4**	**9**	**16**	**25**	**36**
p(a)	**1/6**	**1/6**	**1/6**	**1/6**	**1/6**	**1/6**

E(X) = $1 \times \frac{1}{6} + 4 \times \frac{1}{6} + 9 \times \frac{1}{6} + 16 \times \frac{1}{6} + 25 \times \frac{1}{6} + 36 \times \frac{1}{6} = \mathbf{15.167}$

NOTE: $E(X)^2$ would be $3.5^2 = 12.25$ which is NOT correct.

Quiz 2

1. Assume $E(X) = 5$. What is $E(2X + 5)$?

2. Assume $X \sim Binomial(12,0.25)$. Find $E(X)$.

LUNCH BREAK